Simplified Instructional Computer(SIC) Machine Architecture

SIC Machine Architecture

- The SIC machine architecture depends on the following features:
  - Memory
  - Registers
  - Data Formats
  - Instruction Formats
  - Addressing Modes
  - Instruction Set
  - Input and Output

SIC Machine Architecture

- Memory
  - Memory consists of 8-bit bytes
  - Any 3 consecutive bytes form a word (24 bits)
  - Total of 32768 (2^{15}) bytes in the computer memory

SIC Machine Architecture

- Registers
  - Five 24-bits registers. Their mnemonic, number and use are given in the following table.

Mnemonic	Number	Use
A	0	Accumulator; used for arithmetic operations
X	1	Index register; used for addressing
L	2	Linkage register; contains the return address whenever control transferred to subroutine
PC	8	Program counter; contains the address of the next instruction to be executed.
SW	9	Status word, including Condition code such as $<, \leq, >, \geq, ==$.

SIC Machine Architecture

- Data Formats
  - Integers are stored as 24-bit binary number
  - 2's complement representation for negative values
  - Characters are stored using 8-bit ASCII codes
  - No floating-point hardware on the standard version of SIC

SIC Machine Architecture

- Instruction Formats
  - All machine instructions on the standard version of SIC have the 24-bit format as shown below

Opcode(8)	x	Address (15)

- Addressing Modes
  - There are two addressing modes available, which are as shown in the below table. Parentheses are used to indicate the contents of a register or a memory location.

Mode	Indication	Target address calculation
Direct	x = 0	TA = address
Indexed	x = 1	TA = address + (x)

SIC Machine Architecture

- Instruction Set
  - Load and store registers - LDA, LDX, STA, STX, etc.
  - Integer arithmetic operations - ADD, SUB, MUL, DIV
    - All arithmetic operations involve register A and a word in memory, with the result being left in A
  - COMP – Comparison instruction
  - Conditional jump instructions - JLT, JEQ, JGT
  - Subroutine linkage - JSUB, RSUB
  - I/O (transferring 1 byte at a time to/from the rightmost 8 bits of register A)
    - Test Device instruction (TD)
    - Read Data (RD)
    - Write Data (WD)

SIC/XE Machine Architecture

- The SIC/XE machine architecture depends on the following features:
  - Memory
  - Registers
  - Data Formats
  - Instruction Formats
  - Addressing Modes
  - Instruction Set
  - Input and Output

SIC/XE Machine Architecture

- Memory
  - Memory consists of 8-bit bytes
  - Any 3 consecutive bytes form a word (24 bits)
  - Total of 1 Mb (2^{20}) bytes in the computer memory

SIC/XE Machine Architecture

- Registers
  - There are nine registers; each register is 24 bits in length except floating point register.
  - Their mnemonic, number and uses are shown in the following table.

Mnemonic	Number	Use
A	0	Accumulator; used for arithmetic operations
X	1	Index register; used for addressing
L	2	Linkage register; contains the return address whenever control transferred to subroutine
B	3	Base register; used for addressing
S	4	General working register-no special use
T	5	General working register-no special use
F	6	Floating point accumulator (48 bits)
PC	8	Program counter; contains the address of the next instruction to be executed.
SW	9	Status word, including Condition code such as $<, \leq, >, \geq, =$

SIC/XE Machine Architecture

- Data Formats
  - Integers are stored as 24-bit binary number
  - 2's complement representation for negative values
  - Characters are stored using 8-bit ASCII codes
  - Support 48 bit floating-point numbers

1	11	36
s	exponent	fraction

  - There is a 48-bit floating-point data type, $F*2^{(e-1024)}$

SIC/XE Machine Architecture

- Instruction Formats
- Format 1 (1 byte –

8
opcode

Example: RSUB

- Format 1 (2 byte) -

8	4	4
op	r1	r2

Example: ADDR S, T

- Format 3 (3 byte) -

6	1	1	1	1	1	1	12
op	n	i	x	b	p	e	Displacement

  – Example: LDA #3

- Format 4 (4 byte) -

6	1	1	1	1	1	1	20
op	n	i	x	b	p	e	Address

  – Example: +JSUB RDREC

SIC/XE Machine Architecture

- Addressing Modes and Flag Bits
  - Base relative (n=1, i=1, b=1, p=0)
  - Program-counter relative (n=1, i=1, b=0, p=1)
  - Direct (n=1, i=1, b=0, p=0)
  - Immediate (n=0, i=1, x=0)
  - Indirect (n=1, i=0, x=0)
  - Indexing (both n & i = 0 or 1, x=1)
  - Extended (e=1 for format 4, e=0 for format 3)

SIC/XE Machine Architecture

- Instruction Set
  - Load and store registers - LDA, LDX, STA, STX, LDB, STB etc.
  - Integer arithmetic operations - ADD, SUB, MUL, DIV
  - Floating-point arithmetic operations: ADDF, SUBF, MULF, DIVF
  - COMP – Comparison instruction
  - Conditional jump instructions - JLT, JEQ, JGT
  - Subroutine linkage - JSUB, RSUB
  - Register move instruction: RMO
  - Register-to-register arithmetic operations: ADDR, SUBR, MULR, DIVR

SIC/XE Machine Architecture

- I/O (transferring 1 byte at a time to/from the rightmost 8 bits of register A)
  - Test Device instruction (TD)
  - Read Data (RD)
  - Write Data (WD)